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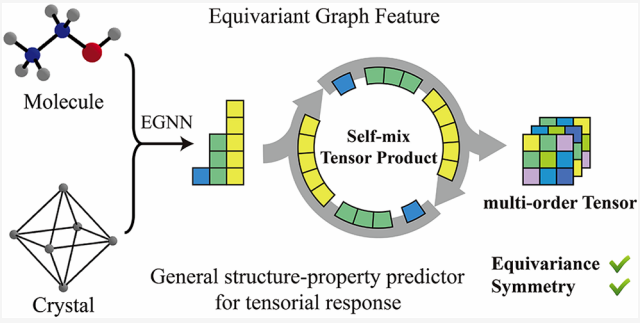
General Framework for Geometric Deep Learning on Tensorial Properties of Molecules and Crystals

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ABSTRACT: Response properties of molecules and crystals are naturally described by tensors that obey specific equivariance and symmetry constraints. However, directly predicting these tensorial quantities remains challenging for machine learning models. We present a general-purpose output module for equivariant graph neural networks that enables end-to-end prediction of tensors of arbitrary order with prescribed permutation (fundamental) symmetry. Coupled with the $SE(3)$-equivariant XPaiNN architecture, our framework attains accuracy comparable to that of first-principles calculations. It also supports atomic-level properties—such as chemical shielding tensors and Born effective charges—in an all-in-one model. Moreover, the method handles higher-order tensors, including molecular hyperpolarizability and the elastic tensor (stiffness matrix) of crystalline materials, thereby enabling the derivation and analysis of rich anisotropic information and facilitating AI-assisted discovery and design of functional molecules and materials.



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INTRODUCTION

The advancement of artificial intelligence (AI) is sparking a surge in science and engineering, while the emerging machine learning (ML) approaches are reshaping the traditional research paradigm of computational science for atomistic systems.[1,2] Building upon the available data sets of interest, it is possible for ML models to exploit relations encoded in training data and predict target properties for unexplored systems with high accuracy, bypassing the routine electronic structure calculations. Given the immense chemical space, ML approaches are valuable and promising solutions for large-scale atomistic simulations,[3,4] screening and rational design of drug molecules,[5] as well as searching for high-performance catalysts[6] and functional materials.[7] In these domains, a vast amount of first-principles computations, mainly through Density Functional Theory (DFT),[8,9] can be impractical or even infeasible.

Respecting the exact geometric and fundamental symmetries of molecules and crystals is regarded as the key to resolving structure−property relationships in atomistic ML methods, which leads to the rise of geometric deep learning.[10] Kernel methods[11] and early deep neural networks (DNNs)[12] often adopt geometric features invariant to translations and rotations as descriptors for the systems. Graph neural networks (GNNs) enhance this representation strategy via message passing,[13] taking advantage of the graph-like nature of molecules and crystals.[14] GNN models operating on invariant geometric feature representations thus demonstrate improved prediction accuracy on benchmark data sets for chemistry and materials science.[15−17] The recent emergence of equivariant GNNs[18−21] enables ML models to resolve symmetry operations of rotation and inversion, achieving the $E(3)$ group equivariance or the $SE(3)$ group equivariance when inversions are not imposed. As a result, equivariant GNNs show significant progress over the invariant models and are employed as surrogates for first-principles calculations in molecular dynamics simulations[22] and materials discovery.[23]

Apart from invariant scalar properties like energy, there are numerous properties of atomic systems that are multiorder equivariant tensors by their response and anisotropic nature. On one hand, these tensorial properties, such as dipole moment $\boldsymbol{\mu}$, polarizability $\boldsymbol{\alpha}$, chemical shielding $\boldsymbol{\sigma}$, dielectric permittivity $\boldsymbol{\epsilon}$, and elasticity $\mathbf{C}$, are directly related to basic observables and phenomena in spectroscopic experiments and mechanics of materials, serving as fundamental ingredients in the description and characterization of functional molecules and materials. On the other hand, the computation of response properties can be significantly more expensive than self-consistent-field (SCF) electronic structure calculations, making high-throughput screening of tensorial targets based on

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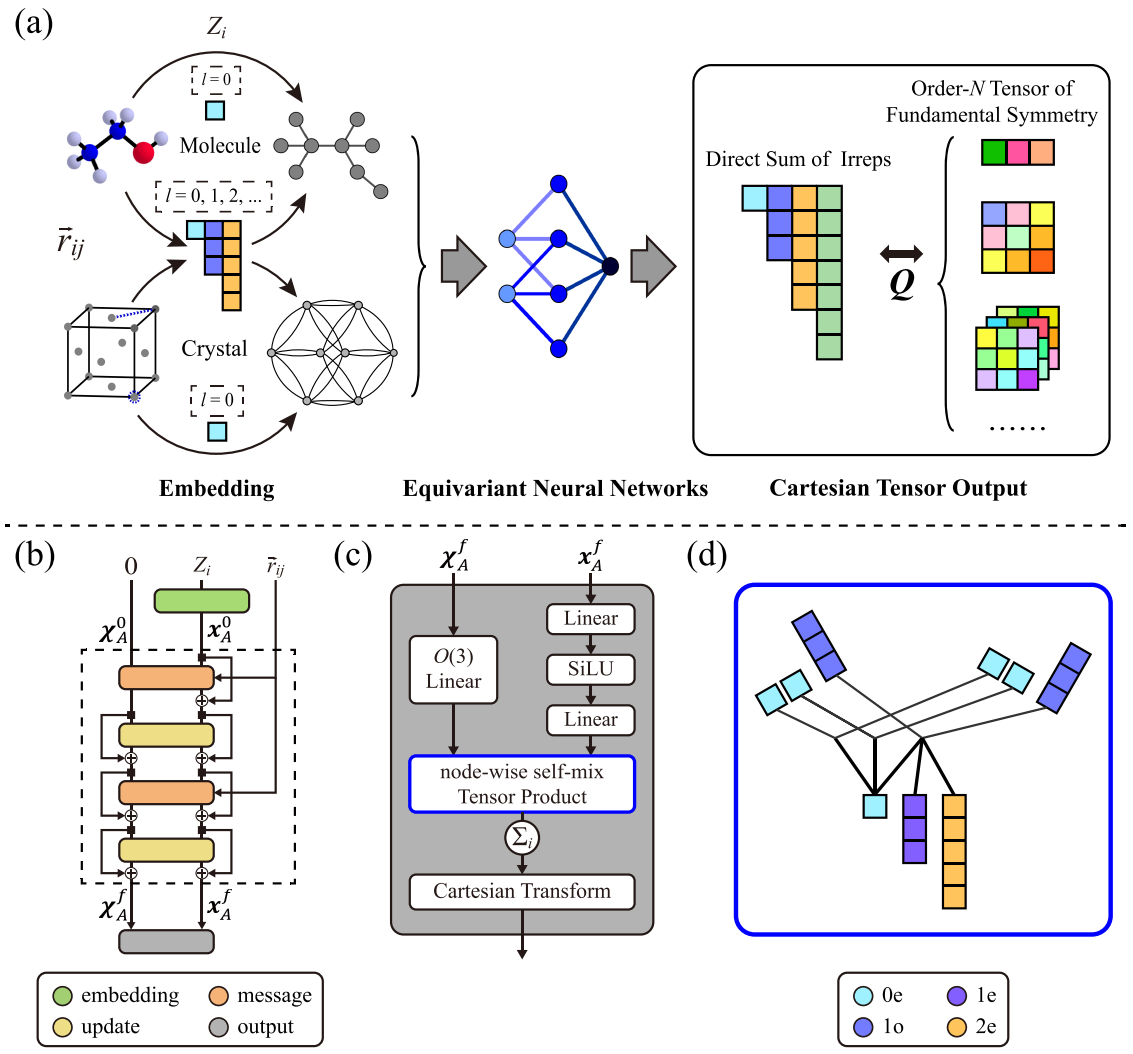


Figure 1. Schematic of learning and predicting the Cartesian tensor with the present XPaiNN model. (a) General workflow of predicting the Cartesian tensor with an equivariant GNN. (b) Model architecture of XPaiNN. (c) Cartesian tensor output module designed in this work, which takes both invariant and equivariant node-hidden representations from the message passing block as input and passes them into the node-wise self-mix layer (marked within the blue frame) with suitable transformations applied. (d) Illustration of node-wise self-mix tensor products, the input $2 \times 0e + 1 \times 1o$ Irreps feature $\boldsymbol{\chi}$ forms the output of $1 \times 0e + 1 \times 1e + 1 \times 2e$, representations for an order-2 tensor of arbitrary symmetry, each line in bold denotes trainable weights transformed from $\boldsymbol{x}$ for each legal tensor product path. The Irreps are given in e3nn[41] notation.

computations more difficult or even impractical. More importantly, tensorial properties, as system response to electric, magnetic, or mechanical perturbations, are more closely correlated with underlying structure and symmetry than scalar properties, resulting in the intrinsic symmetry features of tensors. Thus, developing end-to-end ML methods for learning and predicting tensorial properties of atomistic systems is both important and invaluable.[24]

Some strategies have been proposed for both kernel methods[25,26] and early DNNs[19,27−29] to construct outputs that meet rotation equivariance. Although these strategies are, in principle, feasible with careful design, they have been validated in only a limited number of test cases of single systems. Moreover, most methods are only available for dealing with low-order (order-1 and order-2) tensor outputs and can easily fail for higher-order (order-3 and order-4) targets. This limitation arises because high-order tensorial properties exhibit more complex symmetry constraints than low-order ones. Furthermore, properties such as chemical shielding $\boldsymbol{\sigma}$ and Born effective charges (BEC) $\boldsymbol{Z}^*$ are atom-wise order-2 tensors, representing the local chemical environment, which adds another layer of complexity for existing models and strategies.[24] Several related works have been reported in very recent years,[30−38] offering various strategies for properly constructing tensor outputs. Given their design philosophy, equivariant GNNs are expected to be an ideal framework for predicting tensorial targets. However, satisfying the full symmetry constraints of tensorial targets efficiently poses a significant challenge for equivariant models that directly employ geometric feature representations, especially when predicting the tensorial properties of crystalline systems. In other words, a general and efficient strategy that fully satisfies tensor symmetry across all systems and levels remains elusive, yet it is of paramount importance in geometric deep learning.

In this work, we present a general module for equivariant GNNs that is capable of handling tensor outputs of arbitrary order and fundamental symmetry in the Cartesian form (permutation symmetry of indices). The module is also well suited for predicting atom-wise properties in an "all-in-one" manner. Although the module is mostly showcased here in this

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work to fit in with our recently developed network architecture, XPaiNN,[39] we also demonstrate its flexibility and generality to be adapted to other existing equivariant GNNs. We train and test our model on readily available data sets of both molecules and crystals to demonstrate its capabilities. Leading performances of this model in predicting molecular and atomic chemical properties are observed and analyzed. Additionally, challenging symmetry issues in crystalline systems are reported, validated, and discussed, highlighting the ability of our approach to construct tensorial properties, including pseudotensor components, directly from $SE(3)$-equivariant graph representations composed solely of polar tensors, while accurately capturing the intrinsic symmetry of Cartesian tensors across all seven crystalline systems. Moreover, materials screening via the predicted order-4 elastic tensor is carried out on the Materials Project database,[40] demonstrating the power of the present strategy, capable of finding superhard materials not labeled by DFT calculations.

RESULTS AND DISCUSSION

Equivariant Graph Neural Networks. In atomistic ML, equivariant GNNs operate on geometric tensors constructed using the 3D coordinates of the system. On one hand, spherical models,[18] where features are represented by spherical harmonics with varying degrees of angular momentum l, are generally preferred for their easier access to higher-order feature channels and often more flexible design in architecture,[20,21,41] compared to Cartesian models. On the other hand, the tensor product operations, involving Clebsch-Gordan (CG) coefficients for feature coupling, are computationally intensive, which is a main criticism upon these equivariant models.[2] In contrast, certain Cartesian models avoid CG tensor products by transforming features of different orders separately; these approaches achieve efficiency while maintaining equivariance, as demonstrated by PaiNN[19] and ETGNN.[32] These models are commonly regarded as $SE(3)$-equivariant with equivariant features consisting solely of polar tensor components and are sufficient for modeling many physical properties that do not involve pseudovector couplings. See Supporting Information (SI) for details.

The recent design of the XPaiNN model, proposed by our group, aims to construct an accurate yet efficient equivariant GNN model of general purpose.[39] Building upon the original structure of PaiNN, we extended Cartesian vector feature channel $\mathbf{v}$ with spherical feature channel $\boldsymbol{\chi}$ to include higher l-order features. This channel generates sets of equivariant geometric features that transform as the direct sum of irreducible representations (Irreps) $\boldsymbol{\omega}^{(l,p)}$, as is common in a spherical model.

$$\chi_A = \bigoplus_{(l,p)} \omega_A^{(l,p)} \tag{1}$$

Here, A denotes the Ath node in the graph, and p stands for parity that represents the symmetry of inversion. Within the XPaiNN architecture, the parity for each l-order channel is fixed as $p = (-1)^l$. We retained the scalar feature channel $\boldsymbol{x}_A$ of PaiNN, and let it be transformed simultaneously with $\boldsymbol{\chi}_A$, as illustrated in Figure 1(b). By doing so, we preserved the efficiency characteristic of Cartesian models like PaiNN, as no CG tensor product operation is employed in the main body (i.e., message passing and update blocks) of XPaiNN. This design philosophy has gained recognition in the recent development of other equivariant GNNs.[42,43] Another noteworthy point of XPaiNN is that, unlike the commonly used one-hot embedding, we designed a set of element-wise scalar embedding features $\mathbf{e}^{Z_A}$ for node feature initialization in graph data, differentiated by atomic number Z_A.[39] These features are intended to reflect the periodic laws of chemical elements. The current implementation supports elements within the sixth period, excluding the actinide series.

Cartesian Tensor Output. Diverse strategies have been adopted to predict Cartesian tensors with GNNs. PaiNN proposed a scheme for constructing the order-N tensors via multiple tensor products of vectors, and demonstrated order-1 and symmetric order-2 examples.[19] While PaiNN is a node-based GNN, ETGNN adopts an edge-based GNN design that utilizes the pairwise distance vector $\vec{r}_{AB}$ as the basis for its tensor outputs. As a result, ETGNN was shown to be able to construct order-2 tensor of arbitrary symmetry and order-3 tensor with one permutation symmetry.[32] Later, GMTNet presented a model comparison for dielectric tensor prediction, showing the unsatisfactory results of ETGNN.[35] The results of GMTNet demonstrated that satisfying the intrinsic symmetry of tensors across all crystal space group is a much more stringent requirement, while such symmetry constraints are most straightforwardly satisfied by $E(3)$-equivariant models with full geometric representations of the system, as emphasized in GMTNet[35] and other works on modeling tensorial properties of crystalline systems.[38,44] Consequently, GMTNet calculates crystal tensorial properties by utilizing the derivative method according to the definition of the respective response property.[35] This derivative-based method inherently satisfies the exact physical constraints as defined by the respective response property, fulfilling the intrinsic symmetry requirements by construction.[35,37] Despite these advantages, the derivative-based methods also have some limitations, which are mainly reflected in computational efficiency, implementation complexity when analytically used, and numerical stability when applied numerically.

While it has been addressed from some specific perspectives such as crystal class or space group,[34,36] we highlight here that the fundamental mathematical basis behind all these methods is that any tensor with a given permutation symmetry in its Cartesian indices (i.e., the fundamental symmetry) can be mapped to a unique direct sum of Irreps of the $O(3)$ /$SO(3)$ group in its spherical form. This correspondence is known as the symmetry reduction of tensor products.[41] In recognizing the general applicability of this principle, we propose a general approach for constructing Cartesian tensor outputs with arbitrary tensorial properties across various systems from molecules to crystals. Briefly, the spherical form of the target tensor is directly modeled by the output of an equivariant GNN, and the final Cartesian tensor is readily obtained by transforming the basis from the spherical to the Cartesian form, as illustrated in Figure 1(a). In our approach, both equivariance and fundamental symmetry of tensors are inherently satisfied; see Methods section and SI for detailed discussions. In contrast to many existing approaches that invoke CG tensor products at the stage of message passing and node updating, our approach employs explicit tensor products of equivariant features only at the output stage to construct the desired representations of the target tensor. This enables direct end-to-end learning and prediction of arbitrary tensorial properties and is particularly advantageous for the $SE(3)$-equivariant GNNs with hidden features consisting solely of polar tensor representations to predict output tensors that

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Table 1. Model Test Performances (Mean Absolute Error, MAE) on QM9S Molecule Tensorial Properties

property[a]	unit	SchNet[b]	DimeNet++[b]	PaiNN[b]	NequIP[b]	$SE(3)$-Test-mix[c]	DetaNet[b]	XPaiNN-mix[d]
$\boldsymbol{\mu}$	D	0.0790	0.0246	0.0102	0.0124	0.0116	0.0089	**0.0058** (00)
$\boldsymbol{\alpha}$	a_0^3			0.1536	0.1925	0.1410	0.1367	**0.0543** (03)
$\boldsymbol{\beta}$	a.u.				3.3050	2.2128	2.0373	**1.3947** (62)
$\boldsymbol{D}$	D·Å					0.5166	0.3834	**0.2208** (04)
$\boldsymbol{O}$	D·Å^2					8.9775	6.2442	**4.1583** (58)

[a]Dipole moment $\boldsymbol{\mu}$, polarizability $\boldsymbol{\alpha}$, first hyperpolarizability $\boldsymbol{\beta}$, quadrupole moment $\boldsymbol{D}$, and octupole moment $\boldsymbol{O}$. Best results are given in bold. [b]Results are taken from ref 33. [c]A representative $SE(3)$ model combined with the current output module, see the Methods section. [d]Results are averaged over 4 models, with numbers in parentheses indicating the standard deviation of the MAEs.

contain pseudotensor components, i.e., those require full $O(3)$-equivariance.

The core design of this work is the introduction of the **node-wise self-mix layer** in our output module. This design facilitates the modeling for node-wise contributions $\boldsymbol{T}_A$ of the required spherical representations, taking advantage of the generality of tensor product operation.[41] As illustrated in Figure 1(c,d), this layer consists of customized channel-wise tensor products acting on node-hidden output $\boldsymbol{\chi}_A^f$. Each channel in $\boldsymbol{T}_A$ is computed according to

$$\boldsymbol{t}_{A,w}^{(l_o,p_o)} = \sum_{(l_1,p_1)} \sum_{(l_2,p_2)} \sum_{u} a_{A,uuw}^{(l_o,p_o;l_1,p_1;l_2,p_2)} \circ \boldsymbol{\omega}_{A,u}^{f,(l_1,p_1)} \otimes \boldsymbol{\omega}_{A,u}^{f,(l_2,p_2)} \tag{2}$$

where u and w are the channel index for the given Irrep feature $\boldsymbol{\omega}_A^{(l,p)}$ and $\boldsymbol{t}_A^{(l,p)}$ that constitute $\boldsymbol{\chi}_A^f$ and $\boldsymbol{T}_A$, respectively. $a_{A,\mathrm{uuw}}^{(l_o,p_o;l_1,p_1;l_2,p_2)}$ are trainable parameters for each legal path of the tensor product, $(l_1, p_1) \otimes (l_2, p_2) \rightarrow (l_o, p_o)$, following the selection rules of angular momentum and parity. These parameters are responsible for dimension transformations within the tensor product path and thus $\circ$ denotes matrix multiplication. Notably, a also takes index A in our scheme, indicating that these parameters are node-specific attributes. This is accomplished by transforming the scalar channel's output $\boldsymbol{x}_A^f$ via a multilayer perceptron (MLP) to serve as the path-wise weights, as demonstrated in Figure 1. We utilize this design to fully exploit the capabilities of our XPaiNN architecture (depicted in Figure 1(b)) and to model atom-wise properties, based on the belief that the hidden layer output in GNNs resembles the local chemical environment, in line with the locality assumption,[24] which is well-established in atomistic ML.

Obviously, the tensor product operation applied at the output stage effectively decouples the representation constraints between the hidden equivariant features generated by the GNNs and those of the target tensorial properties. Thanks to this decoupling, those $SE(3)$-equivariant models with hidden features consisting of only polar tensors can still be made to learn and predict tensorial properties with full $O(3)$-equivariance. This is possible because pseudotensor components, such as $\boldsymbol{\omega}^{(1,+1)}$ (1e) and $\boldsymbol{\omega}^{(2,-1)}$ (2o), can be generated at the output stage through tensor products between polar tensors under the rules of angular momentum coupling and parity selection. Notably, our approach overcomes some key limitations of existing direct methods for modeling tensorial properties. Our approach, differing from others in the literature, offers a more efficient, general, and expressive framework for constructing various types of tensor targets across atomic, molecular, and crystalline systems. Further theoretical validations of our strategy regarding the fulfillment of equivariance, fundamental symmetry and intrinsic symmetry are provided in the SI, along with more comparisons with other approaches and detailed illustrations for the design of the **node-wise self-mix layer**.

Finally, since XPaiNN is a node-based GNN, the Cartesian tensor output at the graph level (molecule or crystal) is achieved by summing over all nodes in the graph

$$\mathbf{T} = \mathbf{Q} \sum_{A \in \mathcal{G}} \boldsymbol{T}_A \tag{3}$$

where $\mathbf{Q}$ refers to the transformation matrix to the Cartesian basis, which is precomputed based on the fundamental symmetry of the tensor.[41] For some properties, it is better to average over node contributions, while for atom-wise properties, a direct transformation is applied to $\boldsymbol{T}_A$.

Molecular Tensorial Properties. The QM9S data set was constructed based on the well-established benchmark set QM9,[45] which comprises 130 K organic molecules up to 9 heavy atoms. The QM9S data set includes reoptimized molecular geometries and additional spectroscopic properties calculated at the B3LYP/def-TZVP level of theory.[46] Building upon this foundation, DetaNet, an equivariant GNN, was developed to simulate various spectral properties of organic molecules.[33] Here, we use tensorial properties from the QM9S data set to evaluate our model performance on targets at the molecular level. Separate models are trained and tested for each task (see the Methods section).

As is evident from Table 1, our current output module, when combined with XPaiNN (dubbed XPaiNN-mix), shows remarkably improved and highly consistent performances in predicting tensorial properties of molecules, compared to other models in terms of mean absolute errors (MAEs). Invariant GNN models such as SchNet[15] and DimeNet++,[16] as well as equivariant GNN models like PaiNN, have only been utilized in predicting low-order properties, as there are no higher l-order hidden features in their model architectures. Both NequIP[20] and DetaNet can predict hyperpolarizability $\boldsymbol{\beta}$, by explicitly including $\boldsymbol{\omega}^{(3,-1)}$ in their hidden equivariant features. By contrast, thanks to the self-mix tensor product operation shown in eq 2, XPaiNN-mix can now efficiently capture such contributions to the tensor output without requiring higher l-order hidden features (see model hyperparameters in the SI), achieving better accuracy with lower model complexity. This represents a key advantage for our approach, as model efficiency has long been a practical concern for equivariant GNNs, and is strongly influenced by the maximum l-order of hidden features.[2,42] To further demonstrate the feasibility of our strategy, we have implemented a Test Model, which is configured here to represent a common $SE(3)$-equivariant GNN (see Methods section). The results shown in Table 1 for PaiNN and NequIP are taken from the DetaNet study,[33] where PaiNN's results were obtained using its original output

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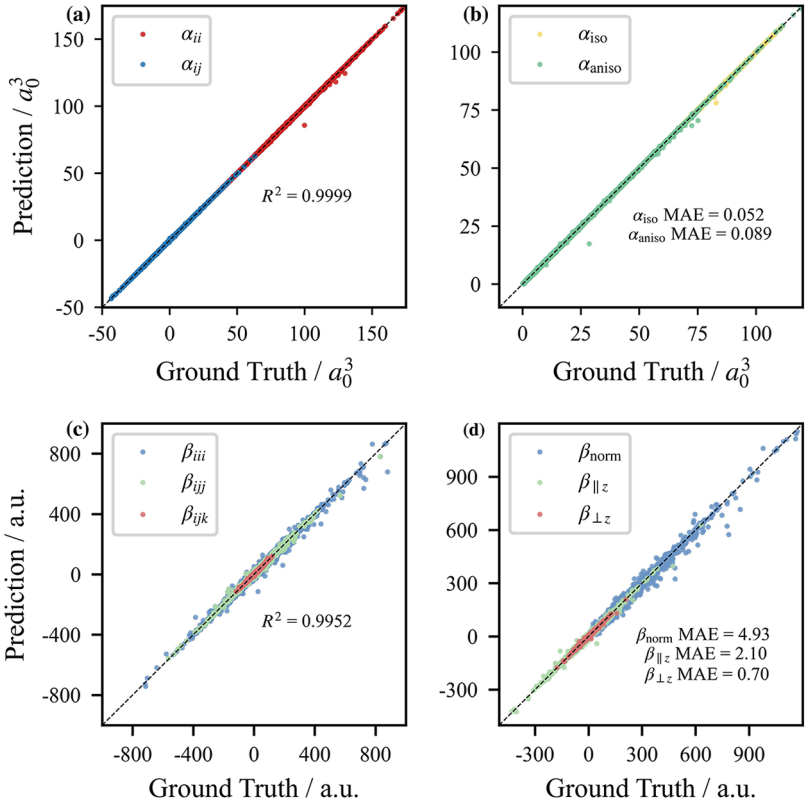


Figure 2. Test performances of XPaiNN-mix on QM9S: (a), (b) predicted polarizability tensor $\boldsymbol{\alpha}$ and the derived isotropic and anisotropic polarizability; (c), (d) predicted hyperpolarizability tensor $\boldsymbol{\beta}$ and the derived magnitude, components parallel and perpendicular to the z axis. Plots of (a) and (c) are presented for the full tensor.

scheme,[19] while NequIP's results were generated using DetaNet's output scheme. In comparison, the $SE(3)$-Test Model, equipped with our output module (dubbed $SE(3)$-Test-mix), achieves competitive performance relative to the results for both PaiNN and NequIP shown in Table 1, demonstrating the broad applicability of our output strategy for equivariant GNNs.

In addition to the excellent performances on the tensors themselves, the XPaiNN-mix model also shows strong predictive power in quantities derived from these tensors. Our model is capable of offering precise predictions of both isotropic and anisotropic polarizability derived from polarizability tensor $\boldsymbol{\alpha}$, as well as accurate estimations of the magnitude and directional components of the first order hyperpolarizability $\boldsymbol{\beta}$. These results are illustrated in Figure 2, underscoring the value of learning and predicting tensorial properties of molecules, especially when anisotropic properties are of interest. More detailed error statistics on the predicted tensors and the derived quantities can be found in the SI.

NMR Shielding Constants and Chemical shifts. Shielding constants and the related chemical shifts are fundamental parameters in nuclear magnetic resonance (NMR) spectroscopy, which are determined as the rotational average over the chemical shielding tensor.

$$\sigma_A = \frac{1}{3}\mathrm{Tr}(\boldsymbol{\sigma}_A) \tag{4}$$

Here, $\boldsymbol{\sigma}_A$ is an atom-wise order-2 tensor of arbitrary symmetry. For atomic properties, conventional ML approaches often use separate models or output transformation layers for different element types of nuclei to enhance the model capacity in classifying the local chemical environment. By contrast, in the current work, we are able to achieve this by using a single model with shared parameters for all atoms. To demonstrate our approach, we use the QM9NMR data set,[47] which contains DFT-calculated shielding constants for the QM9 data set.

Our model achieves an MAE of 0.36 ppm averaged over all nuclei in the test set, which is better than DetaNet's results (0.49 ppm) using 4 separate models trained individually for ^{1}H, ^{13}C, ^{15}N, and ^{17}O, respectively.[33] On a per-element basis, XPaiNN outperforms DetaNet with smaller MAEs of ^{13}C, ^{15}N, and ^{17}O. Notably, the predictions for all heavy atoms are in close agreement with the recently reported NMRNet.[48] While NMRNet is specifically designed for NMR prediction, XPaiNN is able to reach a similar performance with MAE differences within 0.1 ppm for each heavy atom type, as shown in Figure 3(b). Performance on ^{1}H nuclei is slightly inferior to those of DetaNet and NMRNet. This is primarily attributed to the relatively small and narrow numerical distribution of the ^{1}H shielding constants when compared to that of the heavy atoms. As a result, ^{1}H signals contribute less to the overall loss during joint training across all nuclear types, leading to a suboptimal model for ^{1}H. Further detailed comparisons between models are provided in the SI.

To illustrate the capacity of our model in resolving the local chemical environment, we have visualized the hidden features of the scalar channel, $\boldsymbol{x}_A^f$, in XPaiNN using uniform manifold approximation and projection (UMAP).[49] As shown in Figure 3(c,d), for all ^{13}C nuclei sites in the test set, the output features present a clear correlation with the labeled shielding constants. More encouragingly, the results for all ^{15}N and ^{17}O sites not only follow a similar trend but also show clear separation and

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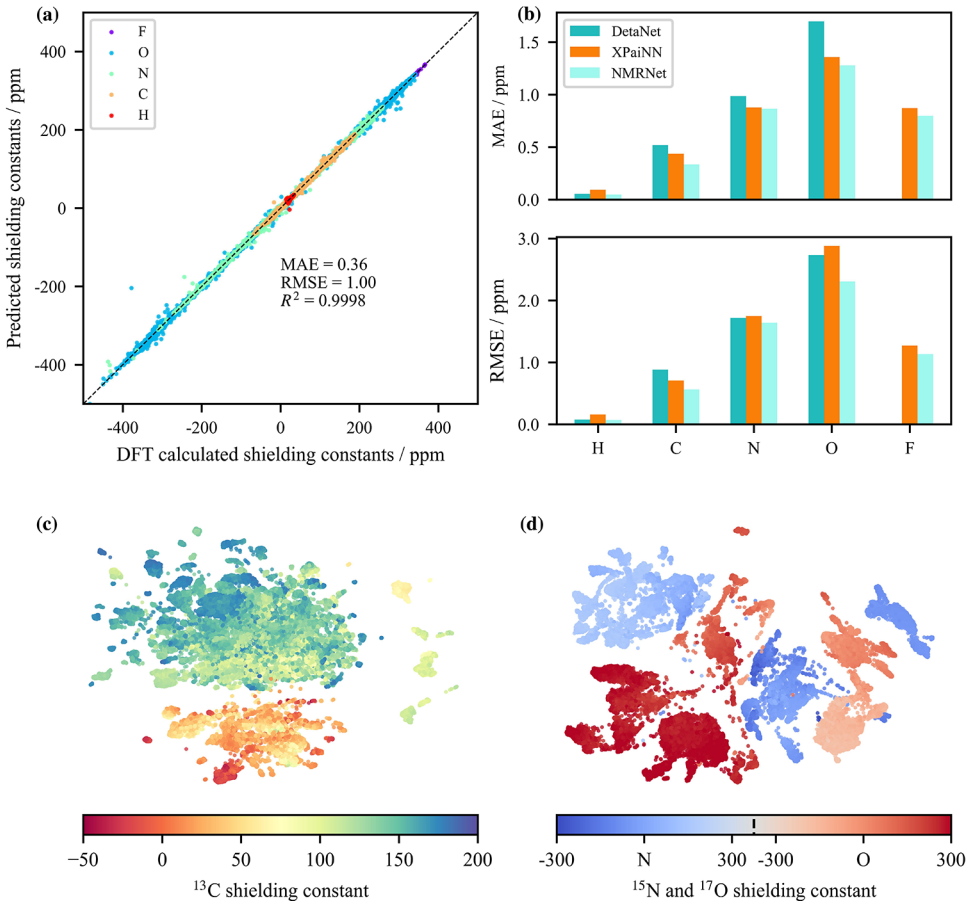


Figure 3. Test performance of XPaiNN on QM9NMR. (a) Element-wise plots of the predicted shielding constants against the ground truth. (b) Comparisons of error statistics for each nuclear type across three models. UMAP (Uniform Manifold Approximation and Projection) analysis of the model's node-hidden features x^f over the test set for (c) all ^{13}C sites and (d) all ^{15}N and ^{17}O sites.

well-defined clustering by element types. This strong structuring in the feature space provides compelling evidence that x_A^f effectively encodes the local chemical environment. These findings firmly support our design of the **node-wise self-mix layer** for modeling the atom-wise target.

To further examine the practicality of our model in learning and predicting atomic properties, we employ the nmrshiftdb2-2024 data set,[48] which contains experimentally measured chemical shifts for diverse molecules, labeled to force field optimized structures. Although it is well-known that reliable chemical shift assignments depend critically on the quality of optimized geometries,[50,51] this data set has emerged as a standard benchmark set for molecular NMR chemical shift predictions. The NMRNet model,[48] built upon the UniMol architecture[52] and pretrained on a large-scale molecular data set, has recently reported state-of-the-art performances on this benchmark set. In contrast to a typical pretraining workflow that relies on supervised learning of atom types and positions, we employ chemical shielding constants as the target. These are readily accessible via quantum chemical calculations and are more closely related to chemical shifts through the definition

$$\delta_A = \sigma^{\mathrm{ref}} - \sigma_A \tag{5}$$

where δ_A denotes the chemical shift of nucleus A and σ^{ref} is the reference shielding constant for the given nucleus type. Equation 5 implies that a model trained on shielding constants can be naturally adapted to predict chemical shifts via simple fine-tuning. This enables a robust transfer from computation-based pretraining to experimental measurements; see the Methods section for details.

Here, we employ a model pretrained on QM9NMR as the base model and illustrate the feasibility of our fine-tuning approach on ^{1}H and ^{13}C chemical shifts against nmrshiftdb2-2024. As shown in Table 2, our fine-tuning approach leads to an effective improvement over direct training results for ^{13}C. For ^{1}H, the performance shows only marginal gains in both

Table 2. Model Performances of Predicting ^{1}H and ^{13}C Chemical Shifts in nmrshiftdb2-2024 via Various Training Strategies

		^{1}H		^{13}C	
model	descriptions	MAE[a]	RMSE[a]	MAE	RMSE
XPaiNN	jointly direct training	0.27	0.50	1.32	2.53
XPaiNN	element-wise direct training	0.24	0.45	1.32	2.36
XPaiNN	jointly fine-tuning from QM9NMR	0.27	0.49	1.25	2.26
XPaiNN	element-wise fine-tuning from QM9NMR	0.23	0.45	1.25	2.25
NMRNet[b]	element-wise fine-tuning from UniMol	0.18	0.40	1.10	2.10

[a]Mean absolute error (MAE) and root mean squared error (RMSE), unit in ppm. [b]Results are taken from ref 48.

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Table 3. Model Performances on Predicting Electric Response Properties of the Curated JARVIS-DFT DFPT Data Set

property	unit	model	Num. Crystal[a]	Fnorm[b] ↓	EwT[b] 25% ↑	EwT 10% ↑	EwT 5% ↑
ϵ		$E(3)$-Test-lin[c]	4508	2.89	71.6%	33.7%	15.3%
		$SE(3)$-Test-mix[c]		2.74	72.3%	39.9%	25.9%
		XPaiNN-mix[c]		**2.47**	79.8%	55.0%	**38.8%**
		GMTNet[d]	4713	3.50	**84.5%**	**57.1%**	27.8%
e	C/m^2	$E(3)$-Test-lin[c]	4884	0.29	52.2%	48.3%	46.4%
		$SE(3)$-Test-mix[c]		0.25	56.0%	48.3%	**46.8%**
		XPaiNN-mix2[c]		**0.24**	**59.9%**	**51.5%**	46.6%
		GMTNet[d]	4998	0.37	49.1%	46.3%	45.7%

[a]Total number of samples used for training and testing the model, see the Methods section for details of data cleaning. [b]Metrics on evaluating model performances, the Frobenius norm distance (Fnorm) and high-quality prediction rate (EwT) follow the definition in ref 35. Best results are given in bold. [c]Models implemented in the current work, the suffixes denote the type of output module combined to the model, "lin" represents the commonly used equivariant linear transform, "mix" stands for the self-mix module depicted in Figure 1(c) and "mix2" refers to the modified version based on "mix" and the changes are illustrated in the SI. [d]Results are taken from ref 35.

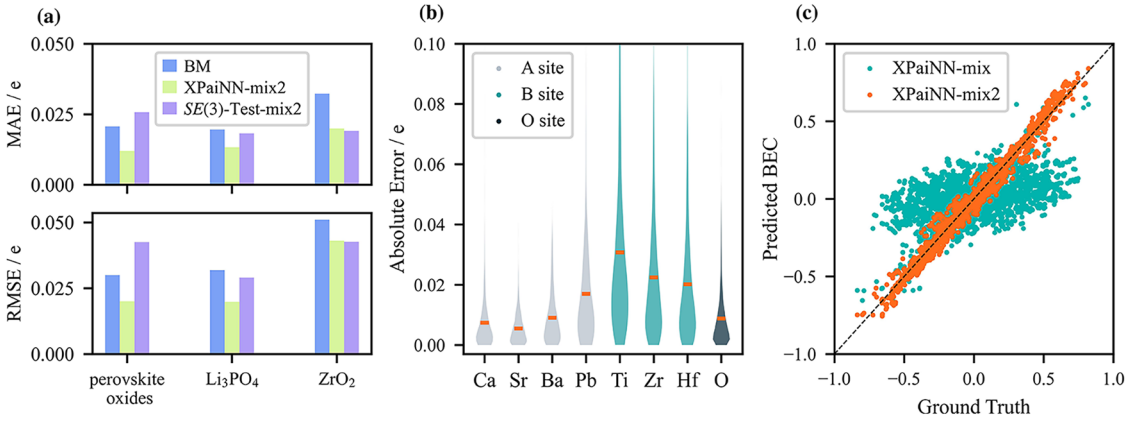


Figure 4. Model performances in predicting Born effective charges. (a) Overall prediction errors for different models across the three BEC data sets. All models are trained on the combined data set, following the same protocol as the baseline model (BM) from ref 38. (b) Element-wise error statistics for XPaiNN-mix2 on the perovskite oxides (ABO_3) data set, shown as violin plots. The orange line indicates the mean absolute error for each element. (c) Comparisons of XPaiNN using the "mix" and "mix2" output modules in predicting the off-diagonal components of the BEC tensor for the central B-site elements (Ti, Zr, and Hf) in perovskite oxides, plotted against the ground truth values.

joint- and element-wise models. This difference in improvement stems from two factors. First, as observed in QM9NMR, the range of ^{1}H chemical shifts is relatively small compared to that of ^{13}C, leading to a less-optimized model in joint training. Second, the predicted shielding constants from the pretrained model do not align well with the target shifts for ^{1}H, partly due to outliers in the data and partly because of the limited chemical diversity of QM9NMR in relation to nmrshiftdb2-2024. We note a performance gap between our model and that of NMRNet. NMRNet benefits from a much larger and more chemically diverse pretraining data set. Nevertheless, the results presented in Table 2 confirm the feasibility of the proposed fine-tuning framework. Performance can be readily improved by employing larger and more diverse data sets for pretraining, particularly since chemical shielding constant data can be efficiently obtained through quantum chemical calculations.

Crystal Electric Response. Response properties are fundamental characteristics of materials that are closely tied to the underlying structure of the crystalline systems. Learning and predicting tensorial properties in crystals remain a significant challenge in satisfying the exact symmetry constraints. Currently, successful approaches have to rely on the full $E(3)$ representation of the system, especially when the target property explicitly contains pseudotensor components. Consequently, computationally intensive CG tensor product convolutions become unavoidable in the message passing blocks of the existing models, such as GMTNet and the related.[35,38]

By representing periodic structures as crystal graphs,[17] our method is inherently capable of learning and predicting the response properties of crystalline materials, benefiting from the self-mix tensor product method introduced in our output module. The DFPT data set from the JARVIS-DFT database[53] contains 5000 crystal structures spanning over 80 elements. These structures are labeled with electric response properties including dielectric tensor $\boldsymbol{\epsilon}$, piezoelectric tensor $\boldsymbol{e}$, and Born effective charge (BEC) tensor $\boldsymbol{Z}^*$, calculated using Density Functional Perturbation Theory (DFPT). While $\boldsymbol{\epsilon}$ is an order-2 symmetric tensor, both $\boldsymbol{Z}^*$ and $\boldsymbol{e}$ are asymmetric or partially symmetric tensors possessing pseudotensor components such as $\boldsymbol{\omega}^{(1,+1)}$ (1e) and $\boldsymbol{\omega}^{(2,-1)}$ (2o), respectively. Moreover, $\boldsymbol{Z}^*$ is an atomic-level property. As noted in ETGNN[32] and GMTNet,[35] the raw DFPT data set contains a considerable number of outliers and structurally distorted systems. To ensure a reliable benchmarking, we use a curated version of the data set provided by GMTNet[35] for evaluating model performances on $\boldsymbol{\epsilon}$ and $\boldsymbol{e}$. For $\boldsymbol{Z}^*$, we employ a recently reported data set[38] with diverse materials including perovskite oxides, Li_3PO_4 and ZrO_2, enabling a more robust assessment of this site-specific tensor prediction.

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Table 4. Model Performances of Predicting Elastic Tensor and Related Elastic Properties in the MatTen Data Set

property[a]	metrics [unit]	$SE(3)$-Test-lin	$SE(3)$-Test-mix	XPaiNN-mix	MatTen[b]	MatSca[b]
$\boldsymbol{C}$	Fnorm ↓[GPa]	41.83	43.95	**31.51**		
	EwT 25% ↑[%]	80.9	81.3	**89.8**		
	EwT 10% ↑[%]	42.9	42.0	**62.2**		
	EwT 5% ↑[%]	16.8	14.4	**31.5**		
K	MAE [GPa]	8.41	8.57	**5.40**	7.37	7.32
	RMSE [GPa]	15.82	14.22	**10.45**		
G	MAE [GPa]	11.05	7.86	**6.33**	8.38	8.63
	RMSE [GPa]	69.58	12.67	**11.23**		
E	MAE [GPa]	22.84	18.50	**14.74**	20.59	19.87
	RMSE [GPa]	71.42	29.78	**26.36**		

[a]Elastic tensor $\boldsymbol{C}$ (in the 6 × 6 stiffness matrix form), bulk modulus K, shear modulus G, and Young's modulus E. Best results are given in bold. [b]Results are taken from ref 34.

To further demonstrate the effectiveness of our approach, we include results from the aforementioned $SE(3)$-Test Model equipped with our output module ($SE(3)$-Test-mix). Additionally, we implement its $E(3)$ version (obtained by configuring the hidden equivariant feature representation of the Test Model, see Methods section) and couple it with the commonly used equivariant linear transformation output,[34,38] forming a baseline model referred to as $E(3)$-Test-lin for comparison. During the experiment, we observed that the current version of our output module ("mix") is suboptimal for predicting tensors with partial symmetry or asymmetric structures such as $\boldsymbol{e}$ (order-3 tensor with 1 permutation symmetry) and $\boldsymbol{Z}^*$ (asymmetric order-2 tensor). This is because, in the "mix" design (Figure 1(c)), the inputs to the node-wise self-mix layer are equivalent, which constrains the expressiveness of the output tensor and limits its ability to model other symmetry patterns. To address this, we slightly modify the "mix" module, introducing an additional equivariant linear transformation layer to construct nonequivalent inputs to the self-mix tensor products, as discussed and illustrated in detail in the SI. This modified version (termed as "mix2") is then used for the prediction of $\boldsymbol{e}$ and $\boldsymbol{Z}^*$.

As shown in Table 3, XPaiNN presents satisfactory results on $\boldsymbol{\epsilon}$ and $\boldsymbol{e}$, being comparable to those of state-of-the-art GMTNet. While the improvement in Fnorm metric is partly related to data cleaning of outliers, results in terms of high-quality prediction rate (EwT) firmly suggest the feasibility of our approach for direct modeling tensorial properties. Notably, the $SE(3)$-mix model outperforms the $E(3)$-lin model across nearly all metrics for both properties, demonstrating the advantages of our output module over the commonly used equivariant linear transformation. A similar trend is observed in the benchmark for $\boldsymbol{Z}^*$, as shown in Figure 4 (a), where the baseline model (BM)[38] resembles the $E(3)$-lin model. As is obvious, when equipped with the "mix2" output module, XPaiNN achieves remarkable improvements over the BM in predicting $\boldsymbol{Z}^*$ across the three systems. Moreover, the $SE(3)$-Test Model also performs better than the BM in Li_3PO_4 and ZrO_2, suggesting that the performance gains in our results arise from a synergistic contribution of both the XPaiNN architecture and the present output module. Furthermore, element-wise analysis of the predicted $\boldsymbol{Z}^*$ for perovskite oxides (ABO_3), depicted in Figure 4 (b), reveals a balanced error distribution across different atomic sites. However, relatively larger errors are observed for the central B-site elements (Ti, Zr, and Hf), which we attribute to relatively small amount of training samples compared to the O- and A-site elements.

Comparison in Figure 4(c) clearly highlights a key limitation of the "mix" module in constructing asymmetric tensors. Specifically, the "mix" model fails to accurately predict the off-diagonal components of the order-2 BEC tensor for central B-site elements, as it cannot capture the antisymmetric contributions from the pseudotensor $\boldsymbol{\omega}^{(1,+1)}$ (1e). By contrast, this issue is effectively resolved by the "mix2" module. By introducing nonequivalent feature input into the self-mix tensor products and, more importantly, the third order polar tensor $\boldsymbol{\omega}^{(3,-1)}$ (3o) feature to the graph representations, the "mix2" module enables the construction of nontrivial pseudotensor components via self-mix tensor products. This is particularly crucial in a highly symmetric local chemical environment, such as the B-site elements in perovskite oxides. Therefore, as demonstrated in Figure 4(c), the predicted off-diagonal tensor components with "mix2" agree well with the ground truth values. These challenges were first discovered in our experiments on the single-crystal systems with perturbed structures,[54] which were employed as case studies in the work of ETGNN.[32] There, we explored a practical solution to address such problems arising from parity ambiguities in $O(3)$-equivariant representations. The current "mix2" design represents a refined solution that resolves these issues while remaining within the framework of polar tensor-based graph representations, ensuring both physical consistency and practical expressiveness. See the SI for further details.

Elastic Tensor of Materials. Elastic tensor $\boldsymbol{C}$, an order-4 tensor with 4-fold permutation symmetry, provides a complete description of materials' response to an external force. Fundamental mechanical properties, such as bulk modulus K, shear modulus G, and Young's modulus E, can be directly derived from $\boldsymbol{C}$. These properties are often employed in characterizing and searching for functional materials. Thus, learning and predicting materials' elastic properties with end-to-end ML models are valuable for understanding and designing materials of intended mechanical properties from the microscopic atomic level.[55−57] Among all these properties, predicting the full elastic tensor is the ultimate goal.[34]

Herein, we train models using XPaiNN and the $SE(3)$-Test architecture, combined either with the "mix" output module or the "lin" module, using the MatTen data set,[58] which contains 10,276 crystals with DFT-calculated elastic tensors. As summarized in Table 4, our XPaiNN model achieves the best performance in all metrics investigated. Model MAEs for predicting bulk and shear moduli are 5.40 and 6.33 GPa, respectively, consistent with the high precision of the predicted full elastic tensor. This leads to the leading performance in

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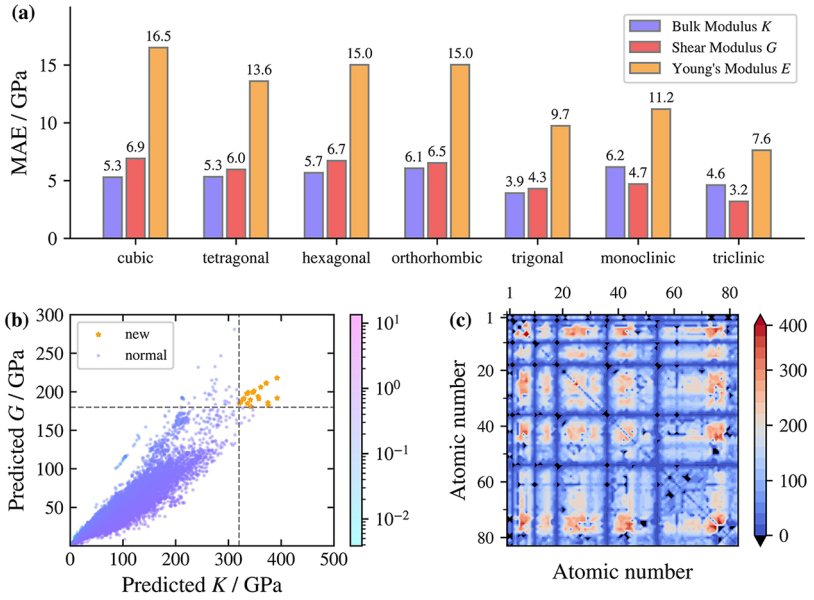


Figure 5. Results of the predicted elastic properties. (a) Test error performances in predicting elastic properties against the MatTen data set, listed by different crystalline systems. (b) Materials screening across 25,021 materials from the Materials Project, plotted by model-predicted shear modulus G vs bulk modulus K. The color shading of the scatter points is determined by the Poisson's ratio calculated from G and K. Potential ultraincompressible and superhard materials identified by the model are located in the upper-right region separated by horizontal and vertical dashed lines and marked with star-shaped symbols. (c) Correlation maps of element combinations, estimated by the maximum K values ever recorded in the pair combinations of elements in binary and ternary compounds. The elements in compounds with unequal atomic fractions are uniformly treated as 1:1. Black areas indicate the missing data in the current data set.

predicting Young's modulus E with an MAE of 14.74 GPa, outperforming other models such as MatTen and MatSca,[34] the latter of which directly fits E without computing the full tensor. To our notice, although performances are quite close between the $SE(3)$-Test-lin and $SE(3)$-Test-mix models on predicting elastic tensor $\mathbf{C}$, the latter yields significantly better results on predicting K, G, and E in terms of RMSE. This suggests that the stiffness matrix converted from the predicted elastic tensor via equivariant linear transformation may be ill-conditioned, leading to large errors in the derived mechanical properties. The related stability phenomena, e.g., occasional negative moduli and nonpositive definite Voigt matrix, were also noted in the work of MatTen,[34] although the RMSE results were not reported for comparison here. In contrast, our "mix" module yields more robust and physically consistent predictions. Furthermore, the best-performing model exhibits balanced error distributions across all seven crystalline systems in the test set, as illustrated in Figure 5(a). This indicates our model's capacity in recognizing crystal symmetry within the $SO(3)$ group, thereby satisfying the intrinsic symmetry requirements of elastic tensor across different crystalline systems. Additional results that support this argument can be found in the SI.

Given the significance of elastic properties in characterizing the mechanical performance of materials, we conducted a materials screening demonstration to explore the usefulness of our ML models in discovering potential ultraincompressible and superhard materials. To eliminate bias from a single-model prediction, we additionally trained another three XPaiNN models with different random seeds and employed an ensemble approach for materials screening. We first validated our method by comparing predictions to recently reported ultraincompressible materials from a study[57] on the OQMD database,[59] achieving good agreement in predicted bulk modulus K. We then applied our models to screen an additional set of 25,021 structures from the Materials Project database.[40] Unlike previous methods,[55,56] we provided predictions of the full elastic tensor, enabling more accurate and physically consistent estimation of mechanical properties. (See the Methods section and the SI for details.)

Our screening criteria identified 20 candidate materials from the full set, as listed in Table 5 and illustrated in Figure 5(b), which are predominantly carbides, borides, and nitrides. Most of them belong to the $P6_3/mmc$ space group, a structural motif commonly associated with superhard materials. Notable among these candidates are previously unreported compounds, such as $Fe(BW)_2$, B_3Os_2, and Mo_2CN, highlighting the predictive power of our model. Furthermore, we conducted a pair-correlation analysis of element combinations among the predicted ultraincompressible materials. As can be seen from Figure 5(c), the potential ultraincompressible materials tend to involve combinations of metals with higher atomic numbers, such as Re/Os, Os/W, and Re/Ir. Meanwhile, combinations of short-period elements (e.g., B, C, and N) with transition metals (e.g., Re, W, and Os) also emerge as promising candidates for ultraincompressible materials. Encouragingly, these findings align well with some of the previously proposed design principles.[56,57] Additionally, we observed that boron (B) can play an important role in the formation of ultraincompressible materials (K > 320 GPa), especially when alloyed with tungsten (W).

CONCLUSION

We developed a general output module for equivariant GNNs, aimed at learning and predicting tensorial properties of both molecules and crystals. Combining this module with our recently proposed $SE(3)$-equivariant GNN model, XPaiNN, we have fully demonstrated in this work that our method is

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Table 5. Potential Superhard Materials Predicted by XPaiNN

ID[a]	formula	K (GPa)[b]	G (GPa)[b]	space group
mp-1019316	Tc_2N	336	185	$P6_3/mmc$
mp-1061298	N_2	392	192	$P6_3/mmc$
mp-1068296	$Fe(BW)_2$	328	191	$Immm$
mp-1070761	$Ni(BW)_2$	321	185	$Immm$
mp-1079348	B_3Os_2	335	198	$P6_3/mmc$
mp-1104181	$Ni(BW)_3$	329	191	$Cmcm$
mp-1186060	$MnOs_3$	348	201	$P6_3/mmc$
mp-1186374	Os_3W	371	211	$P6_3/mmc$
mp-1186902	Re_3Ir	361	206	$Pm\bar{3}$ m
mp-1186923	Re_3Rh	324	187	$P6_3/mmc$
mp-1219509	ReOs	376	182	$P\bar{6}$ m2
mp-1219533	ReIr	356	194	$P\bar{6}$ m2
mp-1220023	OsRu	346	199	$P\bar{6}$ m2
mp-1221519	Mo_2CN	342	190	$Pmm2$
mp-15671	Re_3B	358	191	$Cmcm$
mp-867141	$ReOs_3$	392	218	$P6_3/mmc$
mp-867264	Re_3Os	375	186	$P6_3/mmc$
mp-867345	$TcIr_3$	342	181	$P6_3/mmc$
mp-974358	Ir_3Ru	337	199	$I4/mmm$
mp-974437	Re_2C	372	211	$P6_3/mmc$

[a]Identification in Materials Project database. [b]Model-predicted bulk modulus and shear modulus, averaged over 4 models.

capable of constructing Cartesian tensor outputs of arbitrary order while preserving their fundamental symmetry. This enables $SE(3)$-equivariant GNNs with hidden features consisting solely of polar tensor representations to predict output tensors containing pseudotensor contributions. Furthermore, the intrinsic symmetry of these tensors across different symmetry groups can be naturally satisfied, given that a complete geometric representation can be learned by the GNN model. A critical factor in achieving this is the inclusion of nontrivial polar tensor features with odd parity, which enables the model to properly capture inversion symmetry. By design, our approach is also suitable for predicting atom-wise targets. With minor adjustments, the module can be extended to other properties of interest, such as atomic charges and electron density.[60] Taken together, our strategy is general and well suited for learning and predicting the physical and chemical properties of molecules and crystalline materials. Given the simplicity and effectiveness of the XPaiNN architecture, we anticipate that further advancements in equivariant modeling, when combined with our output approach, can significantly advance the field, potentially enabling the development of foundation models for both molecules and materials.

Response properties of atomistic systems are closely related to the functional properties of molecules and materials. These properties, on one hand, directly determine experimentally measured spectroscopic features; on the other hand, they play important roles in atomistic simulations beyond single-point energy calculations. Therefore, predicting or modeling such properties via ML methods is valuable and offers many practical applications.[37,61] Our current XPaiNN model has not yet accounted for external fields, although many properties are inherently field-dependent. Building on the current strategy, future work should focus on developing a graph representation that can reflect the full system symmetry under applied external fields, a direction as being demonstrated in a related study.[30] Given such a full symmetry graph representation, the output module designed in this work does not encounter any additional complications under an external field.

Capturing fundamental symmetries of atomic systems and their associated physical properties is central to atomistic ML of equivariant quantities.[10,24] In particular, the intrinsic symmetry of the tensorial properties is tightly linked to the geometric symmetry of the systems, especially in the case of crystalline systems. In our current strategy, we do not explicitly impose these symmetry constraints on the output tensors. Instead, we rely on the model's ability to learn them from input graph features, with success hinging on the incorporation of correct physical priors within our architectural design. We believe that learning and predicting tensorial properties serve as a powerful benchmark for evaluating a model's ability to distinguish and encode geometric symmetries in atomistic systems. This remains an active area of research in crystal ML,[35,62,63] yet it is often overlooked in favor of scalar property prediction. Our work contributes to advancing this important research direction.

METHODS

Tensorial Properties Investigated. The various tensorial properties of molecules and crystals exhibit fundamental symmetries due to their nature as response properties. Table 6 summarizes all of the properties investigated in the current work, illustrating that a given symmetry can be mapped to a unique representation in the spherical basis. This is known as the symmetry reduction of tensor products.[41] In the implementation, our model took the tensor order and fundamental symmetry requirement as an input. The final output Irreps and the matrix of basis transformation, $\mathbf{Q}$, shown in eq 3 were then determined by the neural network. This approach is general and

Table 6. Summary of Tensorial Properties of Molecules and Crystals Investigated in This Work

property	symbol[a]	order	fundamental symmetry	Irreps[b]
dipole moment	$\boldsymbol{\mu}$	1	i	1 × 1o
polarizability	$\boldsymbol{\alpha}$	2	$ij = ji$	1 × 0e + 1 × 2e
hyperpolarizability	$\boldsymbol{\beta}$	3	$ijk = ikj = kij$	1 × 1o + 1 × 3o
quadrupole moment	$\boldsymbol{D}$	2	$ij = ji$	1 × 0e + 1 × 2e
octupole moment	$\boldsymbol{O}$	3	$ijk = ikj = kij$	1 × 1o + 1 × 3o
chemical shielding	$\boldsymbol{\sigma}$	2	ij	1 × 0e + 1 × 1e + 1 × 2e
dielectric permittivity	$\boldsymbol{\epsilon}$	2	$ij = ji$	1 × 0e + 1 × 2e
piezoelectricity	$\boldsymbol{e}$	3	$ijk = ikj$	2 × 1o + 1 × 2o + 1 × 3o
Born effective charge	$\boldsymbol{Z}^*$	2	ij	1 × 0e + 1 × 1e + 1 × 2e
elasticity	$\boldsymbol{C}$	4	$ijkl = jikl = ijlk = klij$	2 × 0e + 2 × 2e + 1 × 4e

[a]Symbols used for representing the corresponding tensor in the current work. [b]In e3nn[41] notation.

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applicable to any valid combination of input constraints, not limited to those listed in the table.

Data Set Generation. All data sets used for training and testing in the current work were collected from various sources. Some are publicly accessible, as described in the publications of related works, such as QM9S,[33,46] nmrshiftdb2-2024,[48] the curated JARVIS-DFT,[35] the BEC benchmark,[38] and MatTen.[34,58] The QM9NMR data set is available online. The calculated shielding constants in the gas phase and the geometries relaxed at the B3LYP/6-31G(2df,p) level were taken from ref 64. Examples from ETGNN (single-crystal systems with perturbed structures) can be found at the GitHub repository.[54]

Data cleaning was performed for $\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}$, respectively, in the curated JARVIS-DFT DFPT data set. The cleaning process was based on element composition and thresholds based on the value distributions of the tensorial properties. Further details are provided in the SI, along with the descriptions of model hyperparameters. The 25,021 crystal structures were filtered from the Materials Project database,[40] using the criteria based on element composition and stability ($\Delta E_{\text{hull}} < 10^{-6}$ eV/atom), via the Application Programming Interface (API) wrapped in the MatMiner package[65] (version 0.9.3).

Models and Training. The XPaiNN model was implemented in the XequiNet package (see Data Availability Statement), with primary dependencies on PyTorch 2.0.1,[66] torch-geometric 2.3.1[67] and e3nn 0.5.1.[68] The Test Model implemented in the current work represents a common type of modern equivariant GNN. It employed a simple equivariant message passing convolution, where the hidden node feature was updated as

$$\chi_A^t = \chi_A^{t-1} + \sum_{B\in N(A)} \mathcal{W}^{t-1}(R(r_{AB}))\cdot Y(\hat{r}_{AB}) \otimes \chi_B^{t-1} \tag{6}$$

Here, $R(\cdot)$ denotes the radial basis function (RBF), $Y(\cdot)$ represents the real spherical harmonics (RSH), and $\mathcal{W}^t$ refers to trainable weights at each message passing layer. The embedding layer was shared with XPaiNN. The Test Model architecture can be configured to be either $E(3)$-equivariant or $SE(3)$-equivariant, as the message passing convolution defined in eq 6 is compatible with both symmetries. The choice of equivariance is determined by the transformation properties of the hidden node features, one of the model hyperparameters that decides the desired equivariance. Detailed model hyperparameters are provided in the SI. For the Test Model, the output module in Figure 1(c) took node embedding feature x_A^0 as the input scalar feature. As the main purpose of this Test Model was to demonstrate the generality of the output module designed in this work, other choices, such as using the scalar channel of χ_A^t, which is potentially more physically motivated, were not explored here to maintain simplicity.

For XPaiNN, separate models were trained for each investigated property in QM9S, JARVIS-DFT (including the BEC benchmark), and single systems with perturbed structures (details are provided in the SI). A single model was trained for chemical shielding constants of different types of nuclei in QM9NMR. For the fine-tuning of chemical shifts in nmrshiftdb2-2024, our approach requires two parameters for each nuclear type, namely, a slope and an intercept, such that an approximate linear relationship can be established from the shielding constants (predicted by the pretrained model) to the target chemical shifts, according to eq 5. These parameters were determined by linear regression against the training set and were treated as model hyperparameters during fine-tuning (not being optimized) in the current work. The results are provided in the SI. Four models were trained on MatTen with the same data split (as provided) but different parameter initializations. Error statistics were reported according to the best model in the main text. All of the models were trained using two RTX3090 GPUs. Model hyperparameters and detailed instructions are provided in the SI.

Materials Screening. The elastic properties were calculated from the model-predicted full elastic tensors (see SI). The results were then averaged over predictions from the four trained models. Materials with $K > 320$ and $G > 180$ GPa were considered potential superhard candidates. The criteria were determined based on the previous studies[55,56] and verified through comparisons with recently reported ultraincompressible materials from the OQMD database (see the SI for details).

■ ASSOCIATED CONTENT

Data Availability Statement

The data sets and trained models described in this work are available at https://zenodo.org/records/13332260, where the detailed results for materials screening of elastic properties are also provided for interested readers. A lightweight version of the XequiNet package, used to support the models and reproduce the results in this work, is available on GitHub: https://github.com/wjyan39/XequiNet_tensor.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c12428.

Detailed introduction of the method, additional results for further supporting the current work, and training details of the models (Experimental Section) (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Butler, K. T.; Davies, D. W.; Cartwright, H.; Isayev, O.; Walsh, A. Machine learning for molecular and materials science. *Nature* **2018**, *559*, 547−555.

(2) Zhang, X.et al. Artificial Intelligence for Science in Quantum, Atomistic, and Continuum Systems. 2023, arXiv:2307.08423. arXiv.org e-Print archive. https://arxiv.org/abs/2307.08423 DOI: 10.48550/arXiv.2307.08423.

(3) Unke, O. T.; Chmiela, S.; Sauceda, H. E.; Gastegger, M.; Poltavsky, I.; Schütt, K. T.; Tkatchenko, A.; Müller, K.-R. Machine Learning Force Fields. *Chem. Rev.* **2021**, *121*, 10142−10186.

(4) Friederich, P.; Häse, F.; Proppe, J.; Aspuru-Guzik, A. Machine-learned potentials for next-generation matter simulations. *Nat. Mater.* **2021**, *20*, 750−761.

(5) Staszak, M.; Staszak, K.; Wieszczycka, K.; Bajek, A.; Roszkowski, K.; Tylkowski, B. Machine learning in drug design: Use of artificial intelligence to explore the chemical structure-biological activity relationship. *WIREs Comput. Mol. Sci.* **2022**, *12*, No. e1568.

(6) Mai, H.; Le, T. C.; Chen, D.; Winkler, D. A.; Caruso, R. A. Machine Learning for Electrocatalyst and Photocatalyst Design and Discovery. *Chem. Rev.* **2022**, *122*, 13478−13515.

(7) Stanev, V.; Choudhary, K.; Kusne, A. G.; Paglione, J.; Takeuchi, I. Artificial intelligence for search and discovery of quantum materials. *Commun. Mater.* **2021**, *2*, No. 105.

(8) Hohenberg, P.; Kohn, W. Inhomogeneous electron gas. *Phys. Rev.* **1964**, *136*, B864−B871.

(9) Kohn, W.; Sham, L. J. Self-consistent equations including exchange and correlation effects. *Phys. Rev.* **1965**, *140*, No. A1133.

(10) Atz, K.; Grisoni, F.; Schneider, G. Geometric deep learning on molecular representations. *Nat. Mach Intell* **2021**, *3*, 1023−1032.

(11) Bartók, A. P.; Kondor, R.; Csanyi, G. On representing chemical environments. *Phys. Rev. B* **2013**, *87*, No. 184115.

(12) Zhang, L.; Han, J.; Wang, H.; Car, R.; Weinan, E. Deep potential molecular dynamics: a scalable model with the accuracy of quantum mechanics. *Phys. Rev. Lett.* **2018**, *120*, No. 143001.

(13) Gilmer, J.; Schoenholz, S. S.; Riley, P. F.; Vinyals, O.; Dahl, G. E. Neural message passing for quantum chemistry. 2017, arXiv:1704.01212. arXiv.org e-Print archive. https://arxiv.org/abs/1704.01212. (accessed August 19, 2022).

(14) Reiser, P.; Neubert, M.; Eberhard, A.; Torresi, L.; Zhou, C.; Shao, C.; Metni, H.; van Hoesel, C.; Schopmans, H.; Sommer, T.; Friederich, P. Graph neural networks for materials science and chemistry. *Commun. Mater.* **2022**, *3*, No. 93.

(15) Schütt, K. T.; Sauceda, H. E.; Kindermans, P.-J.; Tkatchenko, A.; Müller, K.-R. SchNet - A deep learning architecture for molecules and materials. *J. Chem. Phys.* **2018**, *148*, No. 241722.

(16) Gasteiger, J.; Giri, S.; Margraf, J. T.; Günnemann, S. Fast and uncertainty-aware directional message passing for non-equilibrium molecules. 2020, arXiv:2011.14115. arXiv.org e-Print archive. https://arxiv.org/abs/2011.14115. (accessed August 19, 2022).

(17) Xie, T.; Grossman, J. C. Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Phys. Rev. Lett.* **2018**, *120*, No. 145301.

(18) Thomas, N.; Smidt, T.; Kearnes, S.; Yang, L.; Li, L.; Kohlhoff, K.; Riley, P. Tensor field networks: Rotation- and translation-equivariant neural networks for 3D point clouds. 2018, arXiv:1802.08219. arXiv.org e-Print archive. https://arxiv.org/abs/1802.08219. (accessed October 13, 2022).

(19) Schütt, K.; Unke, O.; Gastegger, M. Equivariant message passing for the prediction of tensorial properties and molecular spectra. 2021, arXiv:2102.03150. arXiv.org e-Print archive. https://arxiv.org/abs/2102.03150. (accessed November 01, 2022).

(20) Batzner, S.; Musaelian, A.; Sun, L.; Gergio, M.; Mailoa, J. P.; Kornbluth, M.; Molinari, N.; Smidt, T. E.; Kozinsky, B. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nat. Commun.* **2022**, *13*, No. 2453.

(21) Liao, Y.-L.; Smidt, T. Equiformer: Equivariant graph attention transformer for 3D atomistic graphs. 2022, arXiv:2206.11990. arXiv.org e-Print archive. https://arxiv.org/abs/2206.11990. (accessed March 13, 2023).

(22) Musaelian, A.; Batzner, S.; Johansson, A.; Sun, L.; Owen, C. J.; Kornbluth, M.; Kozinsky, B. Learning local equivariant representations for large-scale atomistic dynamics. *Nat. Commun.* **2023**, *14*, No. 579.

(23) Merchant, A.; Batzner, S.; Schoenholz, S. S.; Aykol, M.; Cheon, G.; Cubuk, E. D. Scaling deep learning for materials discovery. *Nature* **2023**, *624*, 80−85.

(24) Fedik, N.; Zubatyuk, R.; Kulichenko, M.; Lubbers, N.; Smith, J. S.; Nebgen, B.; Messerly, R.; Li, Y. W.; Boldyrev, A. I.; Barros, K.; Isayev, O.; Tretiak, S. Extending machine learning beyond interatomic potentials for predicting molecular properties. *Nat. Rev. Chem.* **2022**, *6*, 653−672.

(25) Grisafi, A.; Wilkins, D. M.; Csányi, G.; Ceriotti, M. Symmetry-Adapted Machine Learning for Tensorial Properties of Atomistic Systems. *Phys. Rev. Lett.* **2018**, *120*, No. 036002.

(26) Nguyen, V. H. A.; Lunghi, A. Predicting tensorial molecular properties with equivariant machine learning models. *Phys. Rev. B* **2022**, *105*, No. 165131.

(27) Sommers, G. M.; Andrade, M. F. C.; Zhang, L.; Wang, H.; Car, R. Raman spectrum and polarizability of liquid water from deep neural networks. *Phys. Chem. Chem. Phys.* **2020**, *22*, 10592−10602.

(28) Zhang, Y.; Xia, J.; Jiang, B. Physically motivated recursively embedded atom neural networks: incorporating local completeness and nonlocality. *Phys. Rev. Lett.* **2021**, *127*, No. 156002.

(29) Gastegger, M.; Schütt, K. T.; Müller, K.-R. Machine learning of solvent effects on molecular spectra and reactions. *Chem. Sci.* **2021**, *12*, 11473−11483.

(30) Zhang, Y.; Jiang, B. Universal machine learning for the response of atomistic systems to external fields. *Nat. Commun.* **2023**, *14*, No. 6424.

(31) Venetos, M. C.; Wen, M.; Persson, K. A. Machine Learning Full NMR Chemical Shift Tensors of Silicon Oxides with Equivariant Graph Neural Networks. *J. Phys. Chem. A* **2023**, *127*, 2388−2398.

(32) Zhong, Y.; Yu, H.; Gong, X.; Xiang, H. A General Tensor Prediction Framework Based on Graph Neural Networks. *J. Phys. Chem. Lett.* **2023**, *14*, 6339−6348.

(33) Zou, Z.; Zhang, Y.; Liang, L.; Wei, M.; Leng, J.; Jiang, J.; Luo, Y.; Hu, W. A deep learning model for predicting selected organic molecular spectra. *Nat. Comput. Sci.* **2023**, *3*, 957−964.
(34) Wen, M.; Horton, M. K.; Munro, J. M.; Huck, P.; Persson, K. A. An equivariant graph neural network for the elasticity tensors of all seven crystal systems. *Digital Discovery* **2024**, *3*, 869−882.
(35) Yan, K.; Saxton, A.; Qian, X.; Qian, X.; Ji, S. Symmetry Informed Network for O(3) Equivariant Crystal Tensor Prediction. 2024, arXiv:2406.12888. arXiv.org e-Printarchive. https://arxiv.org/abs/2406.12888. (accessed February 15,2025).
(36) Heilman, A.; Schlesinger, C.; Yan, Q. Equivariant Graph Neural Networks for Prediction of Tensor Material Properties of Crystals. 2024, arXiv:2406.03563.arXiv.org e-Print archive. https://arxiv.org/abs/2406.03563. (accessed February 20, 2025).
(37) Falletta, S.; Cepellotti, A.; Johansson, A.; Tan, C. W.; Descoteaux, M. L.; Musaelian, A.; Owen, C. J.; Kozinsky, B. Unified differentiable learning of electric response. *Nat. Commun.* **2025**, *16*, No. 4031.
(38) Kutana, A.; Shimizu, K.; Watanabe, S.; Asahi, R. Representing Born effective charges with equivariant graph convolutional neural networks. *Sci. Rep.* **2025**, *15*, No. 16719.
(39) Chen, Y.; Yan, W.; Wang, Z.; Wu, J.; Xu, X. Constructing accurate and efficient general-purpose atomistic machine learning model with transferable accuracy for quantum chemistry. *J. Chem. Theory Comput.* **2024**, *20*, 9500−9511.
(40) Jain, A.; Ong, S.; Hautier, G.; Chen, W.; Richards, W.; Dacek, S.; Cholia, S.; Gunter, D.; Skinner, D.; Ceder, G.; Persson, K. Commentary: The Materials Project: A materials genome approach to accelerating materials innovation. *APL Mater.* **2013**, *1*, No. 011002.
(41) Geiger, M.; Smidt, T. e3nn: Euclidean Neural Networks. 2022, arXiv:2207.09453. arXiv.org e-Print archive. https://arxiv.org/abs/2207.09453. (accessed march 10, 2024).
(42) Frank, J. T.; Unke, O. T.; Müller, K.-R.; Chmiela, S. A Euclidean transformer for fast and stable machine learned force fields. *Nat. Commun.* **2024**, *15*, No. 6539.
(43) Yang, Z.; Wang, X.; Li, Y.; Lv, Q.; Chen, C. Y.-C.; Shen, L. Efficient equivariant model for machine learning interatomic potentials. *npj Comput. Mater.* **2025**, *11*, No. 49.
(44) Zhong, Y.; Yu, H.; Su, M.; Gong, X.; Xiang, H. Transferable equivariant graph neural networks for the Hamiltonians of molecules and solids. *npj Comput. Mater.* **2023**, *9*, No. 182.
(45) Ramakrishnan, R.; Dral, P. O.; Rupp, M.; Von Lilienfeld, O. A. Quantum chemistry structures and properties of 134 kilo molecules. *Sci. Data* **2014**, *1*, 1−7.
(46) Zou, Z. QM9S dataset 2023, Version 1. https://figshare.com/articles/dataset/QM9S_dataset/24235333. (accessed October 28, 2023).
(47) Gupta, A.; Chakraborty, S.; Ramakrishnan, R. Revving up 13C NMR shielding predictions across chemical space: benchmarks for atoms-in-molecules kernel machine learning with new data for 134 kilo molecules. *Mach. Learn.: Sci. Technol.* **2021**, *2*, No. 035010.
(48) Xu, F.; Guo, W.; Wang, F.; Yao, L.; Wang, H.; Tang, F.; Gao, Z.; Zhang, L.; E, W.; Tian, Z.-Q.; Cheng, J. Toward a unified benchmark and framework for deep learning-based prediction of nuclear magnetic resonance chemical shifts. *Nat. Comput. Sci.* **2025**, *5*, 292−300.
(49) Healy, J.; McInnes, L. Uniform manifold approximation and projection. *Nat. Rev. Methods Primers* **2024**, *4*, No. 82.
(50) Bagno, A.; Rastrelli, F.; Saielli, G. Toward the Complete Prediction of the ^{1}H and ^{13}C NMR Spectra of Complex Organic Molecules by DFT Methods: Application to Natural Substances. *Chem. - Eur. J.* **2006**, *12*, 5514−5525.
(51) Zhang, Y.; Wu, A.; Xu, X.; Yan, Y. Geometric Dependence of the B3LYP-Predicted Magnetic Shieldings and Chemical Shifts. *J. Phys. Chem. A* **2007**, *111*, 9431−9437.
(52) Lu, S.; Gao, Z.; He, D.; Zhang, L.; Ke, G. Data-driven quantum chemical property prediction leveraging 3D conformations with Uni-Mol+. *Nat. Commun.* **2024**, *15*, No. 7104.
(53) Choudhary, K.; Garrity, K. F.; Sharma, V.; Biacchi, A. J.; Hight Walker, A. R.; Tavazza, F. High-throughput density functional perturbation theory and machine learning predictions of infrared, piezoelectric, and dielectric responses. *npj Comput. Mater.* **2020**, *6*, No. 64.
(54) Zhong, Y. ETGNN data 2023 https://github.com/QuantumLab-ZY/ETGNN. (accessed February 14, 2024).
(55) Tehrani, A. M.; Oliynyk, A. O.; Parry, M.; Rizvi, Z.; Couper, S.; Lin, F.; Miyagi, L.; Sparks, T. D.; Brgoch, J. Machine Learning Directed Search for Ultraincompressible, Superhard Materials. *J. Am. Chem. Soc.* **2018**, *140*, 9844−9853.
(56) Zeng, S.; Li, G.; Zhao, Y.; Wang, R.; Ni, J. Machine Learning-Aided Design of Materials with Target Elastic Properties. *J. Phys. Chem. C* **2019**, *123*, 5042−5047.
(57) Jin, R.; Yuan, X.; Gao, E. Atomic stiffness for bulk modulus prediction and high-throughput screening of ultraincompressible crystals. *Nat. Commun.* **2023**, *14*, No. 4258.
(58) Wen, M.; Horton, M. K.; Munro, J. M.; Huck, P.; Persson, K. A. Elasticity tensors of 10276 crystals from DFT computations 2023, Version v1.0.0, http://zenodo.org/records/8190849. (accessed 2024−04−11).
(59) Kirklin, S.; Saal, J. E.; Meredig, B.; Thompson, A.; Doak, J. W.; Aykol, M.; Rühl, S.; Wolverton, C. The Open Quantum Materials Database (OQMD): assessing the accuracy of DFT formation energies. *npj Comput. Mater.* **2015**, *1*, No. 15010.
(60) Jørgensen, P. B.; Bhowmik, A. Equivariant graph neural networks for fast electron density estimation of molecules, liquids, and solids. *npj Comput. Mater.* **2022**, *8*, No. 183.
(61) Wang, J.; Wang, Y.; Zhang, H.; Yang, Z.; Liang, Z.; Shi, J.; Wang, H.-T.; Xing, D.; Sun, J. E(n)-Equivariant cartesian tensor message passing interatomic potential. *Nat. Commun.* **2024**, *15*, No. 7607.
(62) Cheng, J.; Zhang, C.; Dong, L. A geometric-information-enhanced crystal graph network for predicting properties of materials. *Commun. Mater.* **2021**, *2*, No. 92.
(63) Liang, C.; Rouzhahong, Y.; Ye, C.; Li, C.; Wang, B.; Li, H. Material symmetry recognition and property prediction accomplished by crystal capsule representation. *Nat. Commun.* **2023**, *14*, No. 5198.
(64) Gupta, A.; Chakraborty, S.; Ramakrishnan, R. QM9NMR dataset 2021 https://moldis-group.github.io/qm9nmr. (accessed January 19, 2024).
(65) Ward, L.; Dunn, A.; Faghaninia, A.; et al. Matminer: An open source toolkit for materials data mining. *Comput. Mater. Sci.* **2018**, *152*, 60−69.
(66) Paszke, A.et al. PyTorch: An Imperative Style, High-Performance Deep Learning Library. 2019, arXiv:1912.01703. arXiv.org e-Printarchive. https://arxiv.org/abs/1912.01703. (accessed March 03, 2024).
(67) Fey, M.; Lenssen, J. E. Fast Graph Representation Learning with PyTorch Geometric. 2019, arXiv:1903.02428. arXiv.org e-Print archive. https://arxiv.org/abs/1903.02428. (accessed March 10, 2024).
(68) Geiger, M.et al. Euclidean neural networks: e3nn *Zenodo* 2022 DOI: 10.5281/zenodo.6459381. Version 0.5.1. (accessed March 10, 2024).

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